

RAG-Based PDF Chatbot – Project Q&A Documentation

Setup Guide

Prerequisites

- Python 3.8 or higher
- Ollama installed and running (for local LLM inference)
- Required Python packages (see requirements.txt)

Installation Steps

1. Clone the repository
2. Install dependencies:
pip install -r requirements.txt
3. Start the Ollama server with the llama3 model
4. Run the app using Uvicorn:
python -m uvicorn app.main:app --reload --host 0.0.0.0 --port 8000

Tools & Libraries Used

Backend & API

- FastAPI – Modern web framework
- Uvicorn – ASGI server for FastAPI

PDF Handling

- PyPDF2 – Extracts raw text from PDFs

Embeddings & NLP

- sentence-transformers

- Model: paraphrase-multilingual-MiniLM-L12-v2

****Vector Search****

- FAISS – Fast similarity search

- Index type: IndexFlatL2 (Euclidean)

****Other****

- httpx, Pydantic, python-multipart

Sample Queries & Responses

প্রশ্ন: এইচএসসি বাংলা প্রথম পত্রে কী কী বিষয় অন্তর্ভুক্ত আছে?

উত্তর: এইচএসসি বাংলা প্রথম পত্রে অন্তর্ভুক্ত বিষয়গুল

প্রশ্ন: পরীক্ষার কাঠামো কেমন?

উত্তর: পরীক্ষার কাঠাম

API Overview

****1. /chat – Ask a Question****

- Method: POST

- Description: Submit a question and get an answer

- Request:

```
{ "query": "Your question here" }
```

- Response:

```
{ "answer": "Answer based on the PDF" }
```

****2. /upload_pdf – Upload a PDF****

- Method: POST

- Description: Upload a new PDF to process

- Response:

```
{ "status": "PDF uploaded and index updated." }
```

****Interactive Docs****

- Swagger UI: <http://localhost:8000/docs>
- ReDoc: <http://localhost:8000/redoc>

Evaluation Metrics

Metric	Description	Target	Status	
----- ----- ----- -----				
Response Relevance	Do answers make sense based on the query?	> 80%	TBD	
Context Accuracy	Is the retrieved content accurate?	> 85%	TBD	
Multilingual Support Implemented	Handles both Bangla and English	☒ 100%	☒	
Response Time	How fast is the response?	< 5 secs	TBD	
Memory Retention	Maintains chat context	5 turns	☒ Done	

Technical Q&A

1. ****How was text extracted from PDFs?****

- Tool: PyPDF2
- Cleaned with regex, whitespace normalized
- Unicode support ensured for Bengali

2. ****How did you chunk the document?****

- Word-based sliding window (500 words, 50 overlap)
- Maintains meaning and context

3. ****Embedding Model****

- Model: paraphrase-multilingual-MiniLM-L12-v2

- Chosen for speed, multilingual support

4. **Similarity Search**

- FAISS with L2 distance (IndexFlatL2)
- Fast, simple, and effective

5. **Query Matching**

- Embeds query + chunks in same space
- Handles vague input with history + LLM reasoning

6. **Room for Improvement**

- Semantic chunking, reranking, hybrid search, larger corpora

Final Thoughts

The chatbot combines PDF processing, semantic embeddings, vector search, and local LLM inference using Ollama.

It's effective for both Bangla and English educational queries.

Can be improved with more documents, smarter chunking, and hybrid search.